



TECHNICAL SPECIFICATION FOR ATLAS ABP 200 (200 TPH) STATIONARY HOT ASPHALT BATCH PLANT





ATLAS MODEL ABP 200 ASPHALT BATCH MIX PLANT

Atlas Model ABP-200 comes with Computerized panel and Bag house Type Pollution Control Unit - rated capacity 200TPH @ 3% Moisture content.

General Description

Stationary, Batch type, Fully Automatic Asphalt Mixing Plant – Model ABP-200



Plant Performance

200 ton per hour – 2500 Kg Batch with 80 cycles per hour. The capacity is based on mean density of 1.6 ton/m³ and hot mix temperature of 150°C at mixer outlet.

When density is less or hot mix temperature requirement is high or moisture content is high, the production capacity drops proportionately.

The rated capacity is based on moisture content of 3% and other standard output parameters.

Aggregate should be clean and should not be covered with dirt.

Aggregate shall not contain oversized material, which may block the feeding path.

Filler shall contain less than 1% moisture and shall be free from agglomeration.

Plant Operation: Fully Automatic electro-pneumatic operation with Manual over ride controls.

COLD AGGREGATES BIN FEEDER

Feeding Hoppers

Five Nos. Cold aggregates hopper with a heaped capacity of approx. 75 Ton.

Wide opening bin width to accommodate different loader capacities.

Anti Bridging bin design to ensure smooth and easy material flow, irrespective of the material level.

Sturdy and proven Feeder Unit



* 500 mm wide x 2.750m long belt feeders with an adjustable radial discharge gate.

Adjustable skirts are provided to prevent leakage between bin bottom and belt surface.

The individual bins are provided with feeding units comprising of head and tail pulleys, single roll idlers, endless rubber conveyor, screw type belt tensioners, replaceable rubber skirting, no flow indicators, 0.5 HP vibrator on sand bin, and a 2 HP electric motor supplied through a static inverter variable speed controls with transmission through gear reduction unit.

Individual Feeder Feeding capacity	50 Tones / Hour
Connected power	5 x 2 HP

*Maximum feeding capacity - feeders are usually synchronized as per production demands on site.

Note :- The above specifications are standard for 5 Bin Feeder. If the capacity increases, it may vary.

Gathering Conveyor

Feeder frame integrated, 600 mm wide x 32 m long endless, Gathering conveyor with rated feeding capacity of 250 Tph. 7.5 HP squirrel cage electric motor and speed reduction unit. Supplied with trough idlers, roller guides, return rollers and screw-type belt tensioners.

Feeding capacity	250 Tons / Hour
Connected power	7.5 HP (5.5. KW)

Slinger Conveyor

High speed, 600 mm wide x 18m long Slinger conveyor (Throwing Conveyor), powered by a 5.5 kW squirrel cage electric motor and speed reduction unit is provided with trough idlers, roller guides and return rollers, mounted on a rigid structural frame.

Feeding capacity	250 Tons / Hour
Connected power	7.5 HP (5.5 KW)

DRYER DRUM

Counter flow, Cylindrical (2.10m Dia. x 9.0 m long), inclined type Dryer Drum unit with field proven internal flights designed to give bonus drying capacity with change in moisture content. Internal dryer flights are specifically designed and arranged in the dryer drum to obtain higher heating efficiency and homogeneous heating of different grain size aggregates. Hot aggregates temperature measuring thermocouple type probe for measurement of hot aggregates at the dryer outlet. Dryer drum supports comprising of four heavy duty trunion rollers and two thrust rollers on a rigid frame, and equipped with two trunion rings with Friction wheel drive.



Type	Cylindrical inclined counter flow type.
Max. Temperature	150° C at outlet of the drum
Drum diameter	2,100 mm
Drum length	9,000 mm
Drum slope	5 degrees
Connected power	20 HP (15 KW) X 4 NOS

BURNER (Air Automize)

Dryer is fitted with Low Pressure jet burner with 30 HP fan motor and 7.5 HP fuel pump auto on off control.

Blower: 40 HP (22.4 KW)

Fuel Pump: 5 HP

Burner Rating: 500 to 1500 LPM (5 to 15 MW)

BURNER

Model	FCMB/LO/M/1500
Type	Air Automize
Fuel	Diesel, LDO
Capacity	500 to 1500 Lph

TWO STAGE DRY DUST COLLECTION SYSTEM.

(BAG HOUSE FILTER UNIT)

Primary Dust Collecting Unit.

Primary dust collection system in form of a Pre-separator is provided to entrap coarser dust particles from the dryer drum exhaust and scavenging pipes of the batching tower. Coarser dust is feed thru screw conveyor to the hot elevator.

Type	Screw Conveyor
Screw conveyor	7.5 HP (5.5 KW)

High capacity Exhaust fan powered by a 150 HP motor with Variable Frequency Drive.

Type	Turbo Fan with VFD
Connected power	150 HP (112 KW) (VFD) /120 HP

Bag House Filter Unit

The two stage dry dust collection system has a bag house filter with around 618 m² of filtering area. The 480 nos. of bags in Nomex filtering media, operating on the latest low maintenance, high efficiency reverse Air Flow type de-dusting mechanism. Safety Interlocks (eg high exhaust temperature etc.) and controls to ensure the safe functioning of the unit. Fully automatic De-dusting system incorporating 18 nos. of degusting devices operated by pneumatic cylinder.



Pre-separator / cyclone

Maximum cleaning efficiency

Type	Reverse Air Flow type bag filter
Number of bags	480(300 mm x 2000 mm)
Filter area	618 m ² (approx).
Bag material	"Meta Aramid" felt
Normal temperature limit	180oC
Dust emission	0.1 g/m ³ (Normal)
Connected Power	18 pneumatic cylinder control by solenoid valve.
Capacity	51,000 M3/Hr.

Fines Re-cycling System.

Fines recycling system comprising of a screw conveyor for conveying the entrapped fines in the bag house to the filler elevator, which in turn feeds the filler weigh hopper atop the mixing tower

Connected Power	10 HP (7.5 KW)
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BATCHING TOWER



BATCHING TOWER

Hot Elevator

Totally enclosed, vertically mounted, centrifugal discharge type bucket elevator. The 220 TPH hot elevators is powered by an 25 HP motor and is supplied complete with wear plates, electrical / mechanical drives and wiring. The hot elevator is provided with a roll back arresting device.

Type	Centrifugal discharge type bucket Elevator.
Capacity	220 tph
Connected power	25 HP (Inline Silo Capacity)

Vibrating screen

Completely enclosed Five Deck vibrating screen (2300mm wide x 4665mm long), sieves four different aggregate sizes. The Inclined circulating, vibrating type screen is provided with duplex spring absorbers, and is powered by a 6.2 KW x 2 vibrating motor. The inclined circular motion vibrating screen minimizes clogging of screen sieves compared with low head linear motion screens. Supplied along with One set of Sieves with the following sieve sizes:-3x4 mm, 7x7 mm, 16x16 mm, 25x25 and 40x40 mm fitted with the screen.



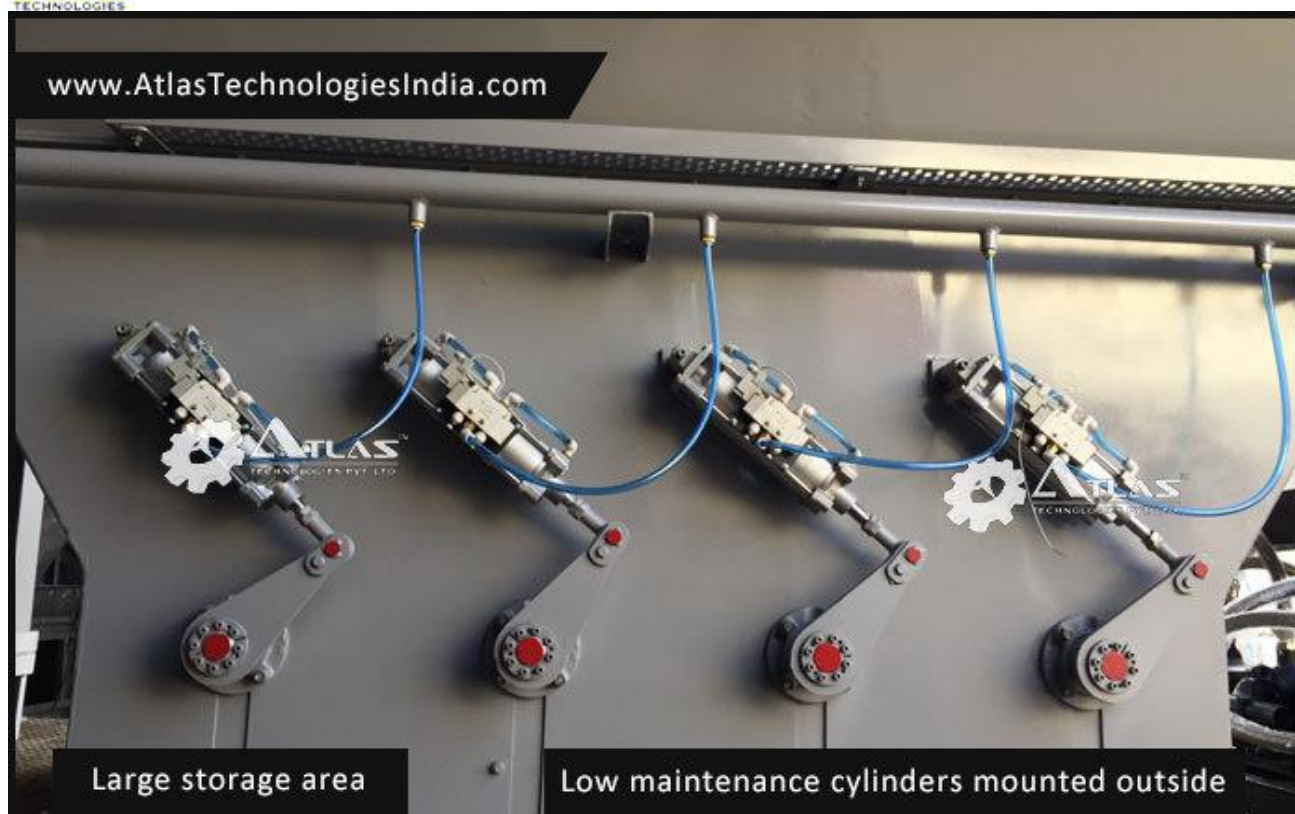
The lower two deck sieves are hooked at longitudinal ends avoiding the need to remove upper deck sieves when replacement of sieves is necessary. Wide platform around screen for easy and safe exchange or adjustment of screen sieves and maintenance.

Type	5 decks, inclined circular vibration type
Capacity	220 tph
Width x Length	1800 mm X 4275 mm
Motor	6.2 KW X 2 Vibrating Motor



Hot Aggregates Bin

Five compartment hot aggregates bin with 25m³ T storage capacity, over flow and over size rejection chute, bin level indicators, pneumatic controlled cut off gates and a sampling device on each compartment. Temperature measuring probe on bin no. 1.



Capacity	25.0 m3 (40 ton) total
Level indicator	One point bin level indicator in each compartment
Temp. Indicator	Thermocouple type temperature sensor in bin no. 1

Weighing System

Automatic accumulating, weighing type Aggregate weigh hopper with 2 Discharge gate and four Load cells. Automatic individual weighing type Asphalt and Filler weigh hopper with Butterfly valve and three 3 load cell (Each). Aggregate weigh hopper, capacity 3000 kg (1.8 m3), Filler weigh hopper, capacity 300 kg (0.4m3) and Asphalt weigh hopper, capacity 225 kg. (0.3m3)

Aggregate Weighing Hopper

Type	Accumulating weighing type with load cell transducer
Capacity	3000 kgs.
Scale hopper capacity	1.8 m3

Filler Weighing Hopper

Type	Individual weighing type with load cell transducer
Capacity	300 kgs.
Scale hopper capacity	0.4 m3



Asphalt Weighing Hopper

Type	Individual weighing type with load cell transducer
Capacity	225 kgs.
Scale hopper capacity	0.3 m3

Pug Mill Type Mixer Unit

Twin shaft, 2500 kg mixing capacity, jacketed pug mill with welded steel shell, lined with replaceable wear resistant spiral liners. Provided with a wide opening slide gate for quick discharge of hot mix. Specially designed arms and tips ensure a very homogeneous mix in a short mixing time. 40 HP x 2 Nos. motors Mixer drive by 2 nos. Shaft mounted Planetary Gear box.

Type	Twin shaft jacketed pug mill
Capacity	2500 Kg / batch
Connected power	40 HP (30 KW) X 2no.
Dimension	2475 X 1550 X 975 mm
Drive	Shaft Mounted
Discharge gate	Pneumatic cylinder operated gate (by 2 nos. Cylinders)
Truck clearance	Height 4000 X 4100 mm Width



FINES / FILLER FEEDING SYSTEM (OPTIONAL)



20M3 filler silo with bucket elevator 5 HP (3.15 KW) motor for feeding filler into filler weighing hopper.
Reclaim filler screw to feed reclaim dust from bag filter 5HP.

ASPHALT PUMPS

Asphalt transfer pump

A 10 HP (7.5 kW) 450 Lpm Jacketed, gear type asphalt transfer pump is provided for feeding asphalt from the storage tank to the asphalt weigh hopper.



Type	Hot oil jacketed gear pump
Pressure	2 Kg/cm ²
Capacity	450 L/min
Motor	10 HP (7.5 KW)

BITUMEN STORAGE AND HEATING SYSTEM

2 nos. indirect heating cylindrical, horizontal asphalt storage tank, capacity 50 Ton. each. Fully insulated 50 MM and clad with internal heat transfer coils and temperature indicator.

Thermic oil heating system with fully automatic light oil burner, hot oil transfer pump, necessary controls and safety interlocks.

Total hot oil transfer piping with necessary control valves. Bitumen jacketed piping with process valves and fittings for manual tank selection and single bitumen delivery pipeline.



Type	Normal pressure, oil heating horizontal tubular Boiler
Capacity	6,00,000 Kcal/Hr input
Fuel	High speed diesel fuel or ASTM D-396 No. 4 fuel oil
Burner	FCMP/LO/S/60
Type	High pressure mechanical atomizing, gun type oil Burner
Capacity	18 to 60 LPH
Control	Automatic ignition, high-low control
Safety controls	Flame eye, low level cut-out float switch, pressure Switch, high temperature cut-out thermostat.
Connected power	1 HP
Circulating pump	
Type	Centrifugal pump
Capacity	30 m3 / hour
Head	55 m
Connected power	10 HP) (7.5 KW)

FUEL TANK

24000 Ltrs capacity, vertical, cylindrical tank for LDO

CONTROL CABIN



Comprising of

Operator's Control Cabin.

2000 mm wide x 4000 mm long, Fully Air Conditioned operator's control cabin complete with weather proof exterior and fitted with tinted toughened glass windows and flush entrance door with lock. Interconnecting wiring (related to all the above sections) designed to suit the positioning of the unit within 10 m radius of the mixing unit.

Electrical Power Control Console

Complete Motor control console for direct online starting and for star delta starting on necessary devices.

Distribution Switch Board

Completely enclosed steel cubicle with push buttons starting of motors on star delta starters.

Magnetic switches with overload thermal relay for each individual motor.

Safety Interlocks and protection devices.

Meters for online current voltage measurements for high capacity motors.

Emergency stop button.

Short circuit protection with switch fuse unit.

Inverter Drives for control of cold aggregates bin feeders.

Temperature Indication.

0 -300°C, Electronic multi point scanner for measuring:

Aggregates temperature measuring in Hot Bin no. 1.

Asphalt feeding temperature.

Bag filter inlet gas temperature.

Bitumen tank temperature. (tank # - 1 & 2)



Sequence Control Panel

- Programmable Logic Control (PLC) with man machine interface.
- Fully automatic electronic solid state sequence control.
- Operator override for manual or semi automatic control.
- Timer controls for both wet and dry cycle.
- Pilot lamps for indication of sequence currently operated.
- Signals and alarms.
- Storage of Fifty different type of mix proportions.
- Automatic free-fall compensation.
- Direct dialing of mix design by percentage.
- Preset batch counter controls the number of batches for a truck-full load.
- Provision for printing operating data on a day to day basis, like - Mix proportions, Batch weight, Total no. of batches, Subtotal, gross total etc

Dryer & Burner Control.

- Hot aggregates temperature at dryer discharge end.
- Remote Burner flame synchronization with hot aggregates temperature.
- Safety interlock, in case of high temperature.

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